$$x_{PB}(t) = x_{BB}(t)\, e^{j2\pi f_c t} + x_{BB}^*(t)\, e^{-j2\pi f_c t}$$

$$= x_A(t) + x_A^*(t)$$

$$\int_{-\infty}^{\infty} x_{PB}(t)\, y_{PB}(t)\, dt = \int [x_A(t) + x_A^*(t)][y_A^*(t) + y_A(t)]\, dt$$

$$= \int x_A(t)\, y_A^*(t)\, dt + \int x_A^*(t)\, y_A(t)\, dt$$

$$+ \int \underbrace{x_A^*(t)\, y_A^*(t) + x_A(t)\, y_A(t)}\, dt$$

$$= 2\,\text{Real}\{\langle x_A, y_A\rangle\} + \int 2\,\text{Real}\{\langle x_A, y_A^*\rangle\}\, dt$$

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A general result $u, v \in L_2$

$$\|u+v\|_2^2 = \langle u+v, u+v\rangle$$

$$= \langle u,u\rangle + \langle u,v\rangle + \langle v,u\rangle + \langle v,v\rangle$$

$$= \|u\|^2 + \|v\|^2 + \langle u,v\rangle + \langle u,v\rangle^*$$

$$= \|u\|_2^2 + \|v\|_2^2 + 2\,\text{Re}\{\langle u,v\rangle\}$$

$\langle x_{PB}, y_{PB}\rangle \Rightarrow$ a real signal

$$\therefore 2\langle x_{PB}, y_{PB}\rangle = 2\,\text{Re}\{\langle x_{PB}, y_{PB}\rangle\}$$

$$= \underbrace{\|x_{PB} + y_{PB}\|^2} - \|x_{PB}\|^2 - \|y_{PB}\|^2$$

$$= 2\|x_A + y_A\|^2 - 2\|x_A\|^2 - 2\|y_A\|^2$$

$$= 2\|x_A\|^2 + 2\|y_A\|^2 + 4\,\text{Re}\{\langle x_A, y_A\rangle\} - 2\|x_A\|^2 - 2\|y_A\|^2$$

$$2\langle x_{PB}, y_{PB}\rangle = 4\,\text{Re}\{\langle x_A, y_A\rangle\}$$

$$\therefore \langle x_{PB}, y_{PB}\rangle = 2\,\text{Re}\{\langle x_A, y_A\rangle\} \;//$$

Or you may use Parseval's

$$
\begin{aligned}
\langle \mathbf{x}_{\mathrm{PB}}, \mathbf{y}_{\mathrm{PB}} \rangle &= \langle \hat{\mathbf{x}}_{\mathrm{PB}}, \hat{\mathbf{y}}_{\mathrm{PB}} \rangle \\
&= \int_{-\infty}^{\infty} \hat{x}_{\mathrm{PB}}(f) \hat{y}^*_{\mathrm{PB}}(f)\, \mathrm{d}f \\
&= \int_{-\infty}^{\infty} \Big(\hat{x}_{\mathrm{A}}(f) + \hat{x}^*_{\mathrm{A}}(-f)\Big)\Big(\hat{y}_{\mathrm{A}}(f) + \hat{y}^*_{\mathrm{A}}(-f)\Big)^*\, \mathrm{d}f \\
&= \int_{-\infty}^{\infty} \Big(\hat{x}_{\mathrm{A}}(f) + \hat{x}^*_{\mathrm{A}}(-f)\Big)\Big(\hat{y}^*_{\mathrm{A}}(f) + \hat{y}_{\mathrm{A}}(-f)\Big)\, \mathrm{d}f \\
&= \int_{-\infty}^{\infty} \hat{x}_{\mathrm{A}}(f)\, \hat{y}^*_{\mathrm{A}}(f)\, \mathrm{d}f + \int_{-\infty}^{\infty} \hat{x}^*_{\mathrm{A}}(-f)\, \hat{y}_{\mathrm{A}}(-f)\, \mathrm{d}f \\
&= \int_{-\infty}^{\infty} \hat{x}_{\mathrm{A}}(f)\, \hat{y}^*_{\mathrm{A}}(f)\, \mathrm{d}f + \left(\int_{-\infty}^{\infty} \hat{x}_{\mathrm{A}}(-f)\, \hat{y}^*_{\mathrm{A}}(-f)\, \mathrm{d}f\right)^* \\
&= \int_{-\infty}^{\infty} \hat{x}_{\mathrm{A}}(f)\, \hat{y}^*_{\mathrm{A}}(f)\, \mathrm{d}f + \left(\int_{-\infty}^{\infty} \hat{x}_{\mathrm{A}}(\tilde{f})\, \hat{y}^*_{\mathrm{A}}(\tilde{f})\, \mathrm{d}\tilde{f}\right)^* \\
&= \langle \hat{\mathbf{x}}_{\mathrm{A}}, \hat{\mathbf{y}}_{\mathrm{A}} \rangle + \langle \hat{\mathbf{x}}_{\mathrm{A}}, \hat{\mathbf{y}}_{\mathrm{A}} \rangle^* \\
&= 2\,\mathrm{Re}\big(\langle \hat{\mathbf{x}}_{\mathrm{A}}, \hat{\mathbf{y}}_{\mathrm{A}} \rangle\big) \\
&= 2\,\mathrm{Re}\big(\langle \mathbf{x}_{\mathrm{A}}, \mathbf{y}_{\mathrm{A}} \rangle\big),
\end{aligned}
$$

$$\|x_{PB}\|_2^2 = 2\|x_A\|_2^2$$

$$\underbrace{\int_0^{\infty} |\hat{x}_{\mathrm{PB}}(f)|^2\, \mathrm{d}f + \int_{-\infty}^{0} |\hat{x}_{\mathrm{PB}}(f)|^2\, \mathrm{d}f}_{} = \|\mathbf{x}_{\mathrm{PB}}\|_2^2 .$$

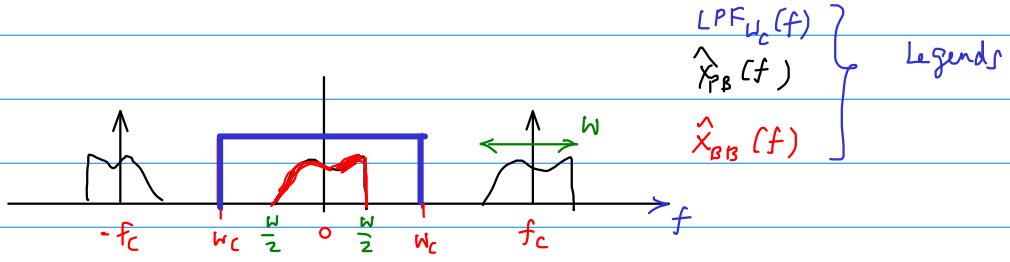
$$= 2\int_0^{\infty} |\hat{x}_{PB}(f)|^2\, df$$

$$\therefore \quad \|x_{PB}\|_2^2 = 2\int_0^{\infty} |\hat{x}_{PB}(f)|^2\, df \qquad -①$$

$$\|x_A\|_2^2 = \int_0^{\infty} |X_A(f)|^2\, df = \int_0^{\infty} |x_{PB}(f)|^2\, df \qquad -②$$

From ① & ② $\quad \|x_{PB}\|_2^2 = 2\|x_A\|^2$ //

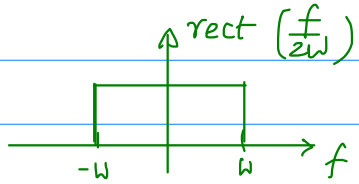
$$x_{BB}(t) = e^{-j2\pi f_c t} x_{PB} * LPF_{W_c}(t)$$



Note: $\frac{W}{2} \leq W_c \leq (f_c - \frac{W}{2})$

$B \leq 2W_c \leq 2f_c - B$

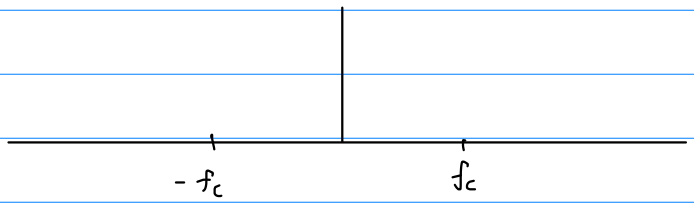
$LPF_{W_c}(t) = 2W_c \, sinc(2W_c t)$ → FT is $rect\left(\frac{f}{2W}\right)$



$|f + 2f_c| \leq \frac{W}{2}$ →

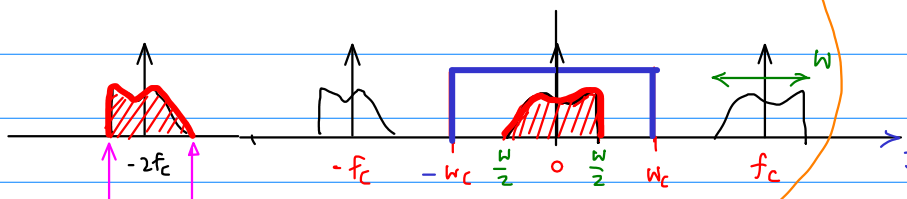
$-\frac{W}{2} \leq f + 2f_c \leq \frac{W}{2}$

$\left(-2f_c - \frac{W}{2}\right) \leq f \leq \left(-2f_c + \frac{W}{2}\right)$



$\frac{W}{2} \leq W_c \leq 2f_c - \frac{W}{2} \Rightarrow W_c \geq \frac{W}{2}$ and $-W_c \geq -2f_c + \frac{W}{2}$

this guarantees that BLUE curve is not in the RED region



filter response must be zero over the region

$|f + 2f_c| \leq \frac{W}{2}$

$$\left.\begin{aligned} g(x) &= f(Ax+b) \\ \nabla g(x) &= A^T \nabla f(Ax+b) \end{aligned}\right\} \quad g(t) = f(x+td) = f(dt+x)$$

$$g'(t) = d^T \nabla f(dt+x)$$

$$g'(0) = d^T \nabla f(x)$$

$$x_{BB}(t) = e^{-j2\pi f_c t} x_A(t)$$

$$y_{BB}(t) = e^{-j2\pi f_c t} y_A(t)$$

$$\langle x_{BB}, y_{BB} \rangle = \int_{-\infty}^{\infty} x_A(t) e^{-j2\pi f_c t} y_A^*(t) e^{j2\pi f_c t} dt$$

$$= \int_{-\infty}^{\infty} x_A(t) y_A^*(t) dt = \langle x_A, y_A \rangle$$

$$y_{PB}(t) = 2\,\mathrm{Re}\{y_{BB} e^{j2\pi f_c t}\} = 2y_{BBR}(t)\cos 2\pi f_c t - 2y_{BBI}(t)\sin 2\pi f_c t$$

$$x_{PB}(t) = 2\,\mathrm{Re}\{x_{BB} e^{j2\pi f_c t}\} = 2x_{BBR}(t)\cos 2\pi f_c t - 2x_{BBI}(t)\sin 2\pi f_c t$$

$$\frac{1}{2}\, y_{PB} * x_{PB}(t) = \left(y_{BBR}(t)\cos 2\pi f_c t - y_{BBI}(t)\sin 2\pi f_c t\right) * \left(x_{BBR}(t)\cos 2\pi f_c t - x_{BBI}(t)\sin 2\pi f_c t\right)$$

$$x_{BB}(t) * y_{BB}(t) = \left(x_{BBR}(t) + j x_{BBI}(t)\right) * \left(y_{BBR}(t) + j y_{BBI}(t)\right)$$

$$= \left(x_{BBR} * y_{BBR}\right)(t) - \left(x_{BBI} * y_{BBI}\right)(t) + j\left(x_{BBR} * y_{BBI}\right)(t) + j\left(x_{BBI} * y_{BBR}\right)(t)$$

appears to be tedious to portray from time-domain

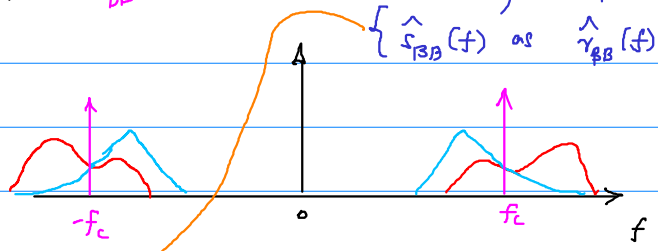
$(x_{PB} * y_{PB})(t) \longleftrightarrow \hat{x}_{PB}(f) \cdot \hat{y}_{PB}(f) = \hat{r}_{PB}(f)$

$(x_{BB} * y_{BB})(t) \longleftrightarrow \hat{x}_{BB}(f) \cdot \hat{y}_{BB}(f) \equiv \hat{s}_{BB}(f)$

$\hat{r}_{PB}(f) = \hat{s}_{BB}(f - f_c) + \hat{s}^*_{BB}(-f - f_c)$

$\Rightarrow$ one may interpret $\hat{s}_{BB}(f)$ as $\hat{r}_{BB}(f)$





that is $\hat{s}_{BB}(f)$ IFT is the

$r_{BB}(t) \equiv (x_{BB} * y_{BB})(t)$

$$x_{\mathrm{BB}}(t) = e^{-i2\pi f_c t} x_{\mathrm{PB}}(t) * \mathrm{LPF}_{W_c}(t)$$

given $x_{PB}(t)$, $x_{BB}(t)$ is the IFT of $\hat{x}_{PB}(f + f_c) \cdot \mathrm{rect}\left(\frac{f}{2W_c}\right)$

analogously

In other words

BB representation of $(x_{PB} * y_{PB})(t)$ is the IFT of shifted (by f_c left) spectrum of $(x_{PB} * y_{PB})(t)$ times

$\mathrm{rect}\left(\frac{f}{2W_c}\right) \cdot \equiv \hat{x}_{BB}(f) \cdot \hat{y}_{BB}(f)$